

DEPARTMENT OF INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Clinical Project

**Prediction of MRI-based Lesion Masks from
CT Perfusion Data**

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Acknowledgments

Abstract

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1 Introduction

Stroke is one of the leading causes of death and disability worldwide, with ischemic stroke accounting for approximately 87% of all cases. In the context of stroke treatment, the phrase "Time is Brain" encapsulates a fundamental principle: time is a crucial factor that significantly influences the extent of irreversible brain damage and the potential for recovery. Each minute delay in treatment results in the loss of approximately 1.9 million neurons, emphasizing the critical nature of rapid intervention.

The challenge in treating ischemic stroke lies in accurately differentiating between irreversibly damaged tissue (the infarct core) and potentially salvageable tissue (the penumbra). As time progresses, the core expands at the expense of the penumbra, highlighting the urgency of accurate diagnostic imaging and prompt treatment decisions.

Currently, computed tomography (CT) is the primary imaging modality used in the acute phase of stroke due to its widespread availability, speed of acquisition, and ability to differentiate between ischemic and hemorrhagic stroke. Perfusion CT, which measures blood flow in the brain over time using contrast agents, plays a crucial role in identifying the core and penumbra regions. However, magnetic resonance imaging (MRI) provides superior soft tissue contrast and more accurate visualization of the final infarct volume, making it the preferred method for follow-up assessment.

This clinical gap—between the initial CT imaging used for urgent decision-making and the subsequent MRI that better represents the actual tissue outcome—presents both a challenge and an opportunity for computational approaches. The ISLES 2024 dataset, the first longitudinal multimodal multi-center real-world dataset in (sub-)acute stroke, offers a unique resource to address this gap by providing paired acute CT and follow-up MRI data from the same patients.

1.1 Project Objectives

The primary goal of this clinical project is to predict MRI-based lesion masks from CT data, effectively bridging the temporal and informational gap between acute CT imaging and follow-up MRI. Specifically, this project aims to:

- Analyze and preprocess imaging data from the ISLES 2024 dataset, including CT perfusion maps (CBF, CBV, MTT, TTP), CT angiography (CTA), CT perfusion

(CTP), and associated clinical information.

- Develop and evaluate computational models that can predict final infarct volumes (as represented in MRI-derived lesion masks) based on sub-acute CT data.
- Assess the generalizability of the developed models across different clinical centers and varying patient characteristics.

The project builds upon recent advances in deep learning approaches to medical image segmentation, particularly in the context of stroke imaging, while addressing the specific challenges posed by variability in imaging parameters across clinical centers.

2 Theoretical Foundations

This chapter provides the theoretical background necessary to understand the clinical and technical aspects of stroke imaging and analysis.

2.1 Ischemic Stroke Pathophysiology

In ischemic stroke, a blood clot obstructs blood flow to a region of the brain, leading to oxygen and nutrient deprivation in the affected area. The brain tissue affected by a stroke can be categorized into two main regions:

- **Infarct Core:** The central region of severely reduced blood flow where brain cells have already died or are irreversibly damaged.
- **Penumbra:** The surrounding area of compromised but still viable brain tissue, which can potentially be saved with timely intervention.

The concept "Time is Brain" emphasizes that as time progresses, the infarct core expands at the expense of the penumbra, making rapid diagnosis and treatment essential to preserve brain function.

2.2 Imaging Modalities in Stroke

2.2.1 Non-Contrast Computed Tomography (NCCT)

NCCT imaging is the first-line imaging modality in acute stroke care due to several advantages:

- Faster acquisition (minutes vs. 30-45 minutes for MRI)
- Widespread availability in hospitals
- No contraindications related to metal implants or pacemakers
- No requirement for patients to remain completely still

- Cost-effectiveness
- Ability to quickly differentiate between ischemic and hemorrhagic stroke

However, CT provides primarily structural and anatomical information and is less sensitive than MRI for detecting early ischemic changes.

2.2.2 Magnetic Resonance Imaging (MRI)

MRI is commonly used for follow-up imaging due to:

- Superior soft tissue contrast
- More accurate visualization of the final infarct volume
- Ability to detect even small ischemic lesions

Diffusion-weighted imaging (DWI) is particularly valuable, as it is the only imaging technique that reliably demonstrates parenchymal injury within minutes to hours from stroke onset.

2.2.3 Computed Tomography Angiography (CTA)

- CT imaging with contrast agents to visualize vascular structures and detect pathologies.

2.2.4 Computed Tomography Perfusion (CTP)

- Dynamic CT method quantifying brain perfusion through temporal contrast distribution analysis.
- Derived perfusion maps:
 - Derived volumetric data visualizing blood perfusion parameters, not raw CT.
 - CBF (Cerebral Blood Flow): Blood flow per time through a brain volume (ml/100g tissue/min).
 - CBV (Cerebral Blood Volume): Total blood volume in a brain region (ml/100g tissue).
 - MTT (Mean Transit Time): Average time for blood to pass through a region.
 - TTP (Time to Peak): Time to maximum contrast agent concentration in an area.

- Tmax (Time to Maximum): Delay until peak of tissue residue function, sensitive indicator for hypoperfusion.

2.2.5 Technical Considerations

Perfusion CT measures blood flow in the brain over time (every 1-2 seconds over 45-60 seconds) using a contrast agent. The derived perfusion maps help differentiate between irreversibly damaged (core) and salvageable (penumbra) tissue. The data used in this project is from two clinical centers: Munich and Zurich. This limited availability of data from multiple centers reduces the variability of deconvolution algorithms and thresholds used in the estimation of core and penumbra volumes.

2.3 Lesion Masks and Ground Truth

Lesion masks represent binary infarct masks in the dataset and are derived from MRI images. They show the actual final infarct areas and are co-registered to the Non-Contrast CT (NCCT). These masks serve as ground truth—the target that predictive models aim to accurately reproduce based on earlier CT data.

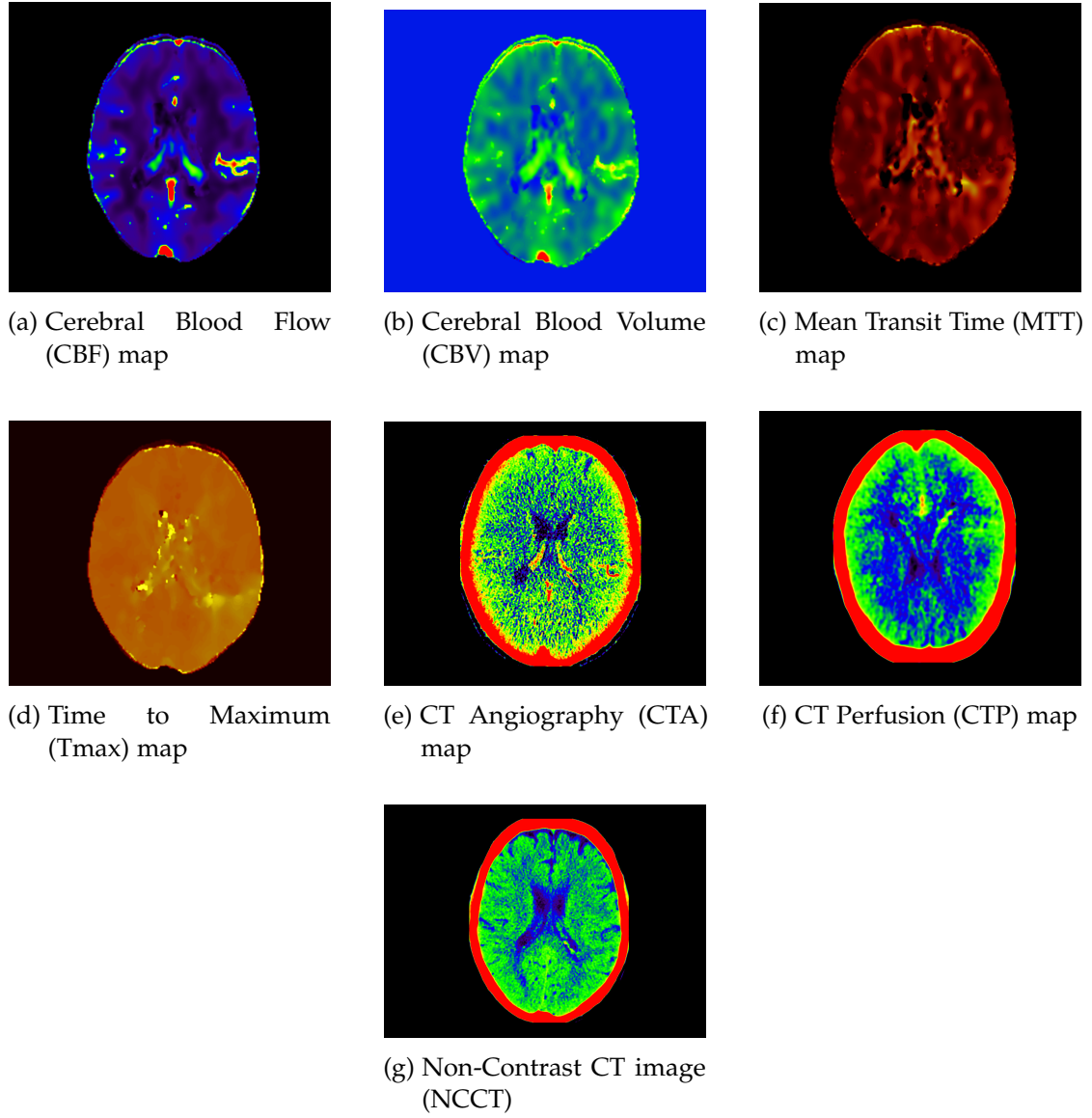


Figure 2.1: Multimodal stroke imaging data

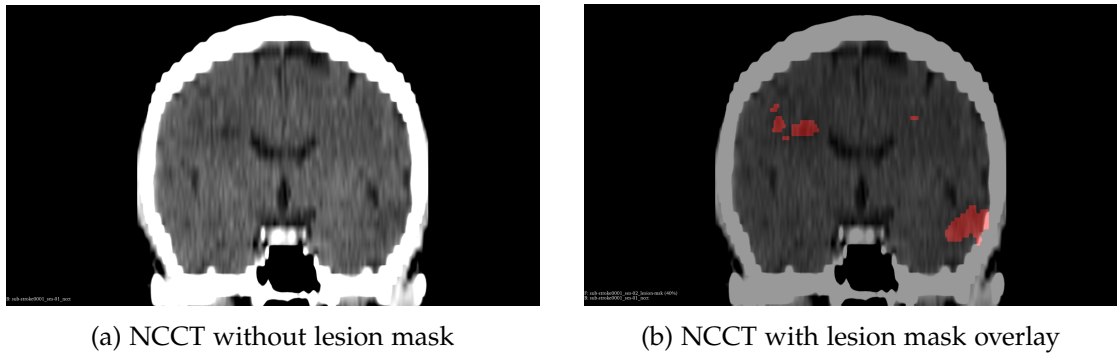


Figure 2.2: Comparison of a Non-Contrast CT (NCCT) image without (left) and with (right) lesion mask overlay. The lesion mask highlights the final infarct area and serves as ground truth for predictive models.

3 Data Preprocessing

3.1 Dataset Structure

The dataset is organized into three main subdirectories: "derivatives", "phenotype", and "raw_data".

- "raw_data" contains the original, un-registered files from the ISLES24 challenge for:
 - ses-01: ncct, cta, ctp, and perfusion maps.
 - ses-02: ADC and DWI MRI files.
- "phenotype" contains clinical and demographic information about the patients.
- "derivatives" contains the processed data for each patient:
 - For ses-01, the processing means registering (aligning) each file to the NCCT file which serves as the base as can be seen from the filename.
 - For ses-02, the processing means generating the label file that is derived from the gold standard DWI.

In the following section, we will visualize the ADC and DWI MRI files for patient SUB-STROKE0001.

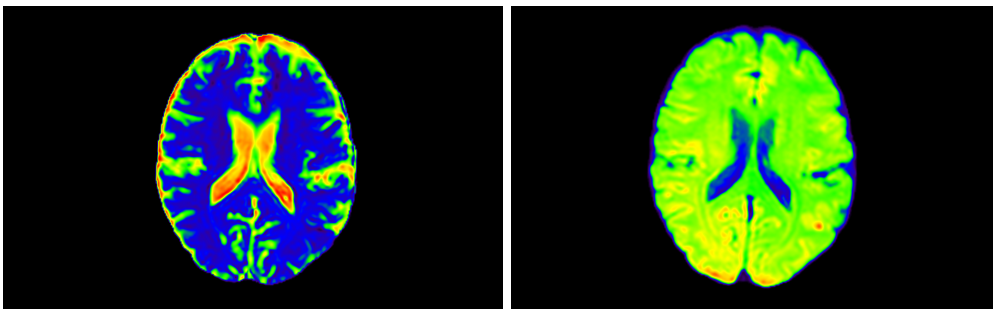


Figure 3.1: Visualized ADC and DWI MRI files for patient SUB-STROKE0001.

The tree subdirectories have a similar structure to the following examples of derivative and raw data for patient SUB-STROKE0001: Since the folder structure is somewhat complex, we use the `tree path /F` command to visualize the structure of one patient's data (stored in SUB-STROKE0001). Each patient in the dataset has a similar directory structure:

```
D:\TUM_CLINICAL_PROJECT\ISLES24_COMBINED\RAW_DATA\SUB-STROKE0001
+---ses-01
|   |   sub-stroke0001_ses-01_cta.nii.gz
|   |   sub-stroke0001_ses-01_ctp.nii.gz
|   |   sub-stroke0001_ses-01_ncct.nii.gz
|   |
|   \---perfusion-maps
|           sub-stroke0001_ses-01_cbf.nii.gz
|           sub-stroke0001_ses-01_cbv.nii.gz
|           sub-stroke0001_ses-01_mtt.nii.gz
|           sub-stroke0001_ses-01_tmax.nii.gz
|
\---ses-02
        sub-stroke0001_ses-02_adc.nii.gz
        sub-stroke0001_ses-02_dwi.nii.gz

D:\TUM_CLINICAL_PROJECT\ISLES24_COMBINED\DERIVATIVES\SUB-STROKE0001
+---ses-01
|   |   sub-stroke0001_ses-01_space-ncct_cta.nii.gz
|   |   sub-stroke0001_ses-01_space-ncct_ctp.nii.gz
|   |
|   \---perfusion-maps
|           sub-stroke0001_ses-01_space-ncct_cbf.nii.gz
|           sub-stroke0001_ses-01_space-ncct_cbv.nii.gz
|           sub-stroke0001_ses-01_space-ncct_mtt.nii.gz
|           sub-stroke0001_ses-01_space-ncct_tmax.nii.gz
|
\---ses-02
        sub-stroke0001_ses-02_lesion-msk.nii.gz
```

This structure follows the BIDS (Brain Imaging Data Structure) format, which is a standardized way of organizing neuroimaging data. The naming convention includes subject ID, session, image space, and modality, making it easier to identify and process the files programmatically.

3.2 Registration

Image registration is a crucial preprocessing step in medical image analysis that aligns different images to a common coordinate system. In the dataset, the registration process ensures that all imaging modalities (CT angiography, CT perfusion, and derived perfusion maps) are spatially aligned to the non-contrast CT (NCCT) image, which serves as the reference space.

Figure 3.2 demonstrates the effectiveness of the registration process by comparing raw and registered Cerebral Blood Volume (CBV) data. The registration successfully aligns the perfusion data to the NCCT reference space, ensuring spatial consistency across all imaging modalities in the dataset.

The derivatives directory contains already registered data for each modality, so we don't have to do the registration by hand.

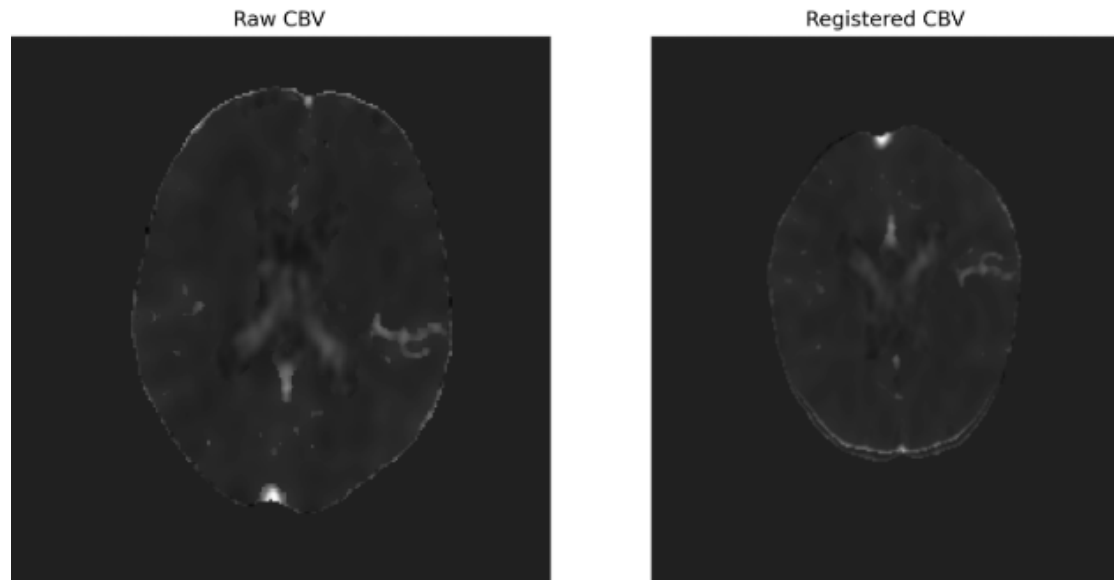


Figure 3.2: Registration comparison for Cerebral Blood Volume (CBV) data. Raw CBV (left), registered CBV (right)

3.3 Normalization

Medical images from different sources can have significant variations in intensity values due to differences in imaging protocols, scanner hardware, and acquisition parameters.

To address this variability and improve the performance of machine learning models, we apply Z-score normalization to our dataset.

3.3.1 Z-Score Normalization

Z-score normalization (also known as standardization) transforms the intensity values of an image to have a mean of 0 and a standard deviation of 1. For each voxel value x in the dataset, the normalization formula is:

$$z = \frac{x - \mu}{\sigma} \quad (3.1)$$

where μ is the mean intensity value of the entire volume and σ is the standard deviation.

This normalization technique offers several advantages for medical image analysis:

- It centers the data distribution around zero, making different scans more comparable
- It helps deep learning models converge faster during training
- It reduces the influence of variations in scanner settings and image acquisition parameters

3.3.2 Normalization Script

To make sure that we work with normalized data, I developed a Python script that iterates over all the NIfTI files in our dataset and applies Z-score normalization to them. Specifically, the script processes all imaging data including CT angiography (CTA), CT perfusion (CTP), and the derived perfusion maps (CBF, CBV, MTT, and Tmax), while excluding lesion masks which should remain as binary data. In addition, the script visualizes the original and normalized images side by side and the difference map. In the following analysis, we examine the results for the first patient (stored in SUB-STROKE0001) in our dataset.

Figure 3.3 demonstrates the effects of Z-score normalization on a CT scan slice. This visualization highlights several important observations about the normalization process:

As clearly illustrated in Figure 3.3, Z-score normalization has several observable effects:

- **Intensity windowing:** Before normalization, we apply a windowing technique (width/level adjustment) to emphasize specific intensity ranges of interest while

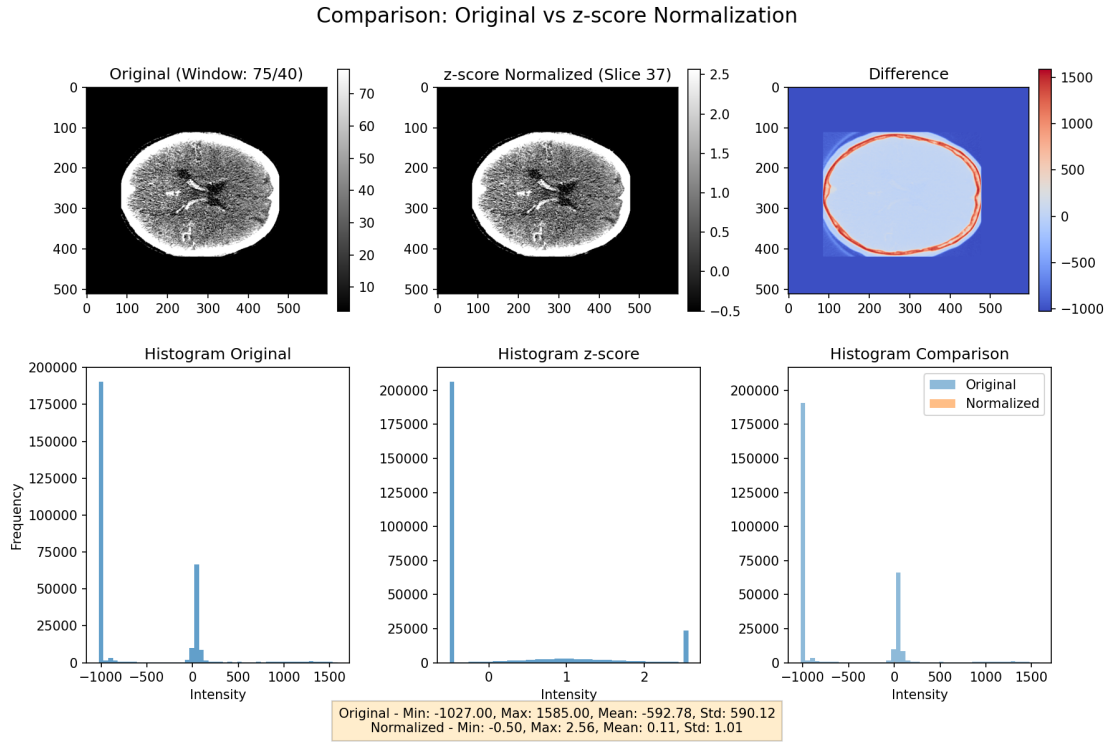


Figure 3.3: Example of Z-score normalization applied to a CT image (slice 37). Top row: Original image (left), normalized image (center), and difference map (right). Bottom row: Corresponding histograms showing intensity distributions before and after normalization. Note the significant change in intensity range and scale after normalization.

de-emphasizing less important values such as those for bone or air. For example, for CTA images we use a window width of 200 and window level of 100, which focuses the normalization on the intensity range most relevant for vascular structures. Different window settings are applied for each imaging modality (CTP, CBF, CBV, MTT, Tmax) to optimize for their specific intensity characteristics.

- **Range transformation:** The original image has intensity values ranging from approximately -1027 to 1585, while the normalized image values are scaled to approximately -0.50 to 2.56. After windowing and normalization, the image values are scaled to approximately -0.42 to 3.62, which demonstrates how the normalization process compresses this wide range into a standardized range centered around zero.
- **Statistical properties:** For this CTA image (slice 37 of patient SUB-STROKE0001), the mean value of the original image is -592.78 with a standard deviation of 590.12. After normalization, the mean is much closer to zero (0.11) with a standard deviation close to one (1.01), which closely aligns with the goal of Z-score normalization.

This standardization process ensures that intensity values across different scans become comparable, which is particularly important when training machine learning models on datasets acquired from different scanners or institutions.

Note: I created this script to work with my local directory structure. If you wish to use this script, please modify the file paths to match your directory structure.

4 UNet Model Implementation

In the following chapter, we will implement a UNet model using the MONAI framework using only one modality, the CTA.

4.1 MONAI Pipeline Overview

The MONAI framework (Medical Open Network for AI) provides a modular pipeline that allows for efficient development of deep learning models in medical imaging. A typical pipeline within MONAI consists of three main stages: data preprocessing, model definition, and training/evaluation. In this project, we employ this structure to implement a UNet model using CTA images as the sole modality. MONAI is specifically designed to empower researchers and clinicians alike by providing domain-specific transformations and workflows for medical imaging tasks.

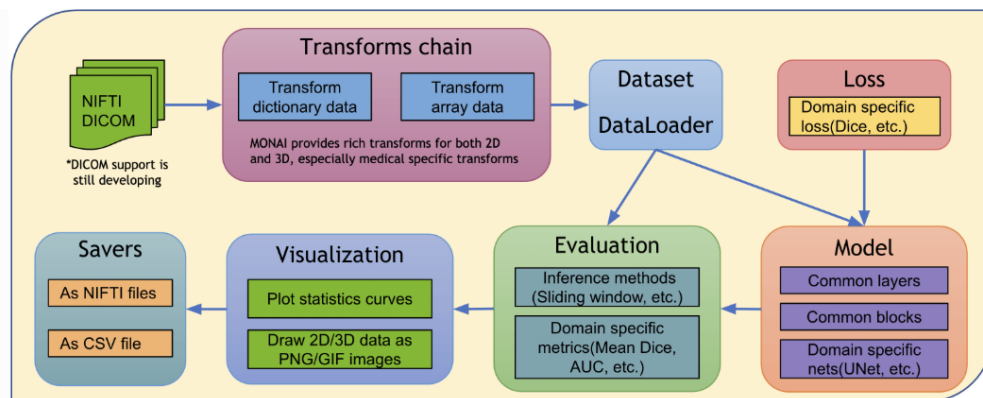


Figure 4.1: Overview of the typical MONAI pipeline for deep learning in medical imaging. The workflow illustrates the complete process from data input (NIFTI files) to the final model. Adapted from [Con20].

Data Preprocessing and Loading

The preprocessing stage ensures that raw CTA scans are transformed into a format suitable for model training. For this, MONAI offers a variety of *transforms*, which can be chained together in a `Compose` object. [Lea23] Key steps in our preprocessing include:

- **Loading and Conversion:** Using `LoadImaged` to read NIfTI files and `EnsureChannelFirstd` to standardize the data layout.
- **Resampling:** Aligning voxel spacing across scans to ensure uniform resolution, which is crucial for batch training.
- **Normalization:** Intensity normalization (e.g., z-score or percentile-based) to reduce inter-patient variability in CTA images.
- **Augmentation:** Random flips, rotations, intensity shifts, and elastic deformations to increase the robustness of the model and reduce overfitting.

Model Definition

Once the data pipeline is established, MONAI allows straightforward definition of deep learning architectures. For example for our UNet model (see Section 4.2) we will use the `monai.networks.nets.UNet`. The parameters of the network such as number of channels can be configured to fit the specific problem of CTA segmentation.

Training and Evaluation

For training, MONAI integrates seamlessly with PyTorch, providing essential components for medical image segmentation. This includes:

- **Loss Functions:** Medical image segmentation often benefits from combinations of Dice loss and cross-entropy loss, which MONAI provides through `monai.losses.DiceCELoss`.
- **Metrics:** MONAI provides medical image-specific evaluation metrics, such as Dice coefficient through `monai.metrics.DiceMetric` and Hausdorff distance through `monai.metrics.HausdorffDistanceMetric`, for rigorous performance assessment.
- **Inference:** MONAI's `monai.inferers.SlidingWindowInferer` enables efficient inference on large volumetric data by processing overlapping patches.

During training, the model leverages MONAI's loss functions and metrics while using standard PyTorch training loops for flexibility and control over the training process.

4.2 UNet Architecture

Unet is a type of convolutional neural network (CNN) that is particularly well-suited for image segmentation tasks [RFB15]. It is a fully convolutional network, meaning that it does not use any fully connected layers. Instead, it uses a series of convolutional layers to extract features from the input image, and then uses a series of upsampling layers to reconstruct the output image. The output of the network is a segmentation mask of the same size as the input image. The UNet architecture is composed of two main parts: the encoder and the decoder. The encoder is a series of convolutional layers that extract features from the input image. The decoder is a series of upsampling layers that reconstruct the output image. You can think of this architecture like downscaling an Image to see the main components like edges or Forms. Because we are interested in these, we should ignore finer details which are abstracted away during downscaling. Connections between the encoder and the decoder are called skip connections. The skip connections allow the network to combine low-level and high-level features, which is crucial for capturing fine-grained details in the output mask.

The UNet architecture is illustrated in Figure 4.2.

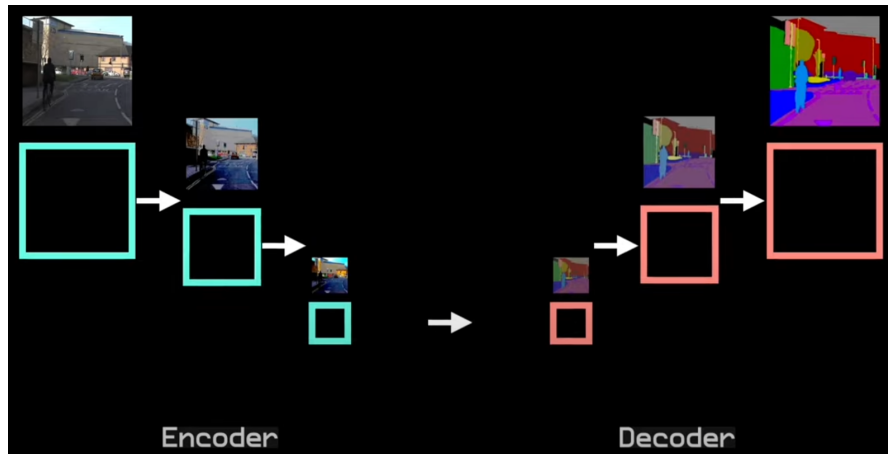


Figure 4.2: Conceptual diagram of the UNet encoder-decoder architecture. The encoder progressively reduces spatial dimensions while extracting higher-level features, while the decoder reconstructs the full-resolution segmentation map. Image adapted from [ai25].

4.3 Implementation

The implementation of the UNet model evolved through several iterations, starting with a minimal viable approach and culminating in a robust pipeline.

4.3.1 Initial Approach: Minimal Viable Model

To start a minimal viable model to establish a baseline and understand the fundamental challenges was implemented. This initial approach used the Binary Cross-Entropy (BCE) loss function and the Dice metric, trained on a single batch for five epochs.

The initial preprocessing pipeline addressed the essential requirements for volumetric data training:

1. **Spatial Consistency:** Unifying voxel spacing to ensure all inputs share the same spatial dimensions
2. **ROI Standardization:** Centering and cropping data to a common region of interest
3. **Synchronized Processing:** Applying transformations simultaneously to both image and mask

Key transformations included:

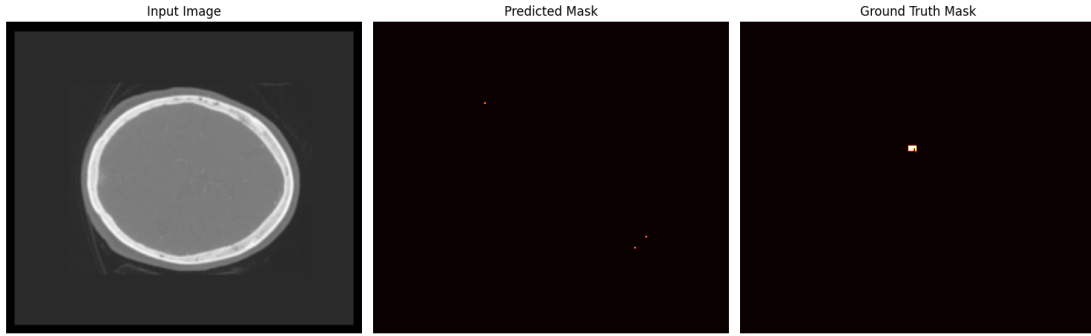
- `Spacingd(keys=["image", "label"], pixdim=[1.0, 1.0, 1.0])` for uniform voxel spacing
- `DivisiblePadd(keys=["image", "label"], k=16)` to ensure voxel shapes are divisible by 16

The training results after five epochs were:

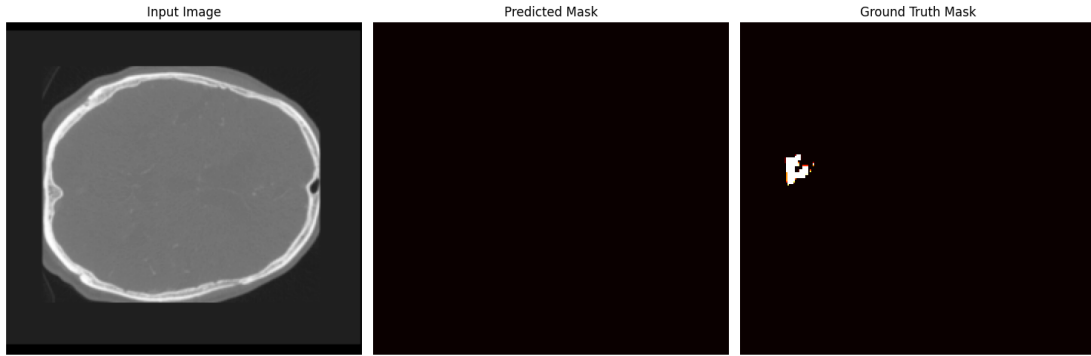
- Loss: 0.1655
- Dice: 0.0000

A Dice score of 0 indicated that the model was not matching the target masks at all, despite the relatively low loss value. This suggested the model was either not detecting relevant features or making entirely incorrect predictions.

As shown in Figure 4.3, the predicted masks did not resemble the ground truth, with the model predicting only background and failing to capture the lesion areas. This failure revealed a fundamental issue with the BCE loss function for this specific problem.



(a) First example of a CT scan slice with corresponding prediction and ground truth mask. The prediction shows no correspondence with the actual lesion.



(b) Second example showing a different CT scan slice. Here too, the model fails to detect the lesion, predicting only background.

Figure 4.3: Comparison between input images (left), predicted masks (middle) and ground truth masks (right) for two different examples. The black prediction masks show that the model mainly predicts the background and fails to detect the lesions.

The BCE loss function proved unsuitable due to severe class imbalance, where the positive class (lesion) occupied only a small portion of the image compared to the background. This imbalance caused:

- **False Negatives:** Small lesions were barely penalized, leading to poor detection
- **Background Bias:** The model learned to detect background features rather than lesion characteristics

With that knowledge the Loss Function had to be changed to the Dice loss function which is more appropriate for this task.

4.3.2 Final Implementation: Robust Pipeline

Building on the lessons learned from the initial approach, the final implementation addresses the identified challenges and incorporates best practices for medical image segmentation.

Configuration and Setup

The implementation uses a centralized configuration approach that allows easy experimentation with different hyperparameters. For the modality, CTA was selected with the window/level settings of 75/40.

Data Preprocessing and Augmentation

The final preprocessing pipeline was significantly enhanced to improve model robustness and training efficiency. The key transformations in the training pipeline are:

Standardization: As before, `LoadImaged`, `EnsureChannelFirstd`, `Orientationd` (to RAS coordinate system, where R=Right, A=Anterior, S=Superior), and `Spacingd` (to $1 \times 1 \times 1$ mm³ voxels) ensure a consistent starting point for all volumes.

Intensity Scaling: For CTA images, a specific `ScaleIntensityRanged` transform was applied, using a window of 75 and a level of 40 Hounsfield Units (HU). This focuses the model on the intensity range most relevant for blood vessels, normalizing the input to a [0, 1] range and clipping other values.

Foreground Cropping: `CropForegroundd` is used to remove large areas of empty background space, allowing the model to focus on relevant anatomy.

Smart Sampling: The most critical addition is `RandCropByPosNegLabeld`. This transform intelligently creates training patches of size (128, 128, 64) by ensuring a balanced sampling of regions containing the lesion (positive samples) and regions containing

only background (negative samples). By setting `pos=2` and `neg=1`, we ensure that for every three patches, two will contain parts of the lesion, which directly combats the class imbalance problem.

Data Augmentation: To prevent overfitting and improve generalization, several random augmentations are applied:

- **RandFlipd:** Randomly flips the patches along any of the three spatial axes.
- **RandRotated:** Applies small random rotations to the patches, simulating variations in patient positioning.

The validation pipeline includes the same standardization and scaling steps but omits the random cropping and augmentation to ensure consistent evaluation on the full, unaltered validation volumes.

Model Architecture and Training Strategy

- **Model:** The core architecture is a 3D `monai.networks.nets.UNet`. It is configured with five encoding levels with channel counts of (32, 64, 128, 256, 512). Crucially, it incorporates residual units and Instance Normalization. Residual connections help mitigate vanishing gradients in deep networks, while instance normalization improves training stability by normalizing feature maps for each sample independently.
- **Loss Function:** The `DiceCELoss` was chosen, which is a compound loss function combining Dice loss and Cross-Entropy loss. This provides the benefits of Dice loss for handling class imbalance while retaining the smooth gradients and per-voxel penalization of Cross-Entropy, leading to more stable training.
- **Optimizer and Scheduler:** The AdamW optimizer is used, which is an improvement over standard Adam that decouples weight decay from the gradient update. This is paired with a learning rate scheduler that implements a 10-epoch linear warmup followed by a cosine decay. This strategy starts with a small learning rate to avoid early training instability, gradually increases it, and then slowly decreases it, which often helps the model converge to a better minimum.
- **Inference:** During validation, the `SlidingWindowInferer` solves a practical problem: how to process large medical volumes that are too big to fit in memory all at once. It works by dividing the large volume into smaller, overlapping patches of

size (128, 128, 64) that the model can handle. Each patch is processed separately, and then the results are intelligently combined to create a complete segmentation map. This approach gives us the best of both worlds: we can train the model on small, manageable patches (which is more efficient and uses less memory), but still segment entire large volumes during inference. Without this technique, we would be limited to only processing volumes the same size as our training patches.

Post-processing

A crucial step in the final pipeline is the post-processing of the model's raw output. Raw predictions from deep learning models often contain small, disconnected regions of "noise" that are clinically irrelevant and can interfere with proper evaluation. The post-processing pipeline refines the binary prediction mask through two main steps:

- **Removing Small Components:** The system analyzes the predicted mask to identify and count separate regions (lesions). Any predicted region smaller than a specified threshold (600 voxels in our implementation) is automatically removed, as these are typically false detections that don't represent real anatomical structures.
- **Morphological Operations:** Two cleaning operations are applied to improve the mask quality. The first operation removes tiny isolated pixels by temporarily shrinking the mask and then expanding it back - this eliminates small false detections while preserving real lesions. The second operation fills small holes within the predicted lesions by temporarily expanding the mask and then shrinking it back - this smooths the boundaries and makes the lesions appear more complete and anatomically realistic.

This post-processing step significantly improves the final segmentation quality by removing artifacts and noise, making the output more clinically plausible and better aligned with evaluation metrics that consider lesion counts and shapes.

4.4 Evaluation Metrics

To comprehensively assess the performance of our UNet model for CTA lesion segmentation, we employ four complementary evaluation metrics that capture different aspects of segmentation quality. These metrics range from voxel-level overlap measurements to object-level detection accuracy, providing a holistic view of the model's capabilities.

The following sections (4.4-4.7) describe each metric in detail. Based on the experimental results from UNet_CTA_fold0, the model demonstrated promising performance with a Dice coefficient of 0.167, though there remains room for improvement in lesion detection accuracy and boundary precision.

4.5 Dice Loss Function

The Dice loss function is commonly used in medical image segmentation tasks due to its effectiveness in handling class imbalance, which is typical in medical imaging where the region of interest (e.g., lesions) often occupies only a small portion of the image.

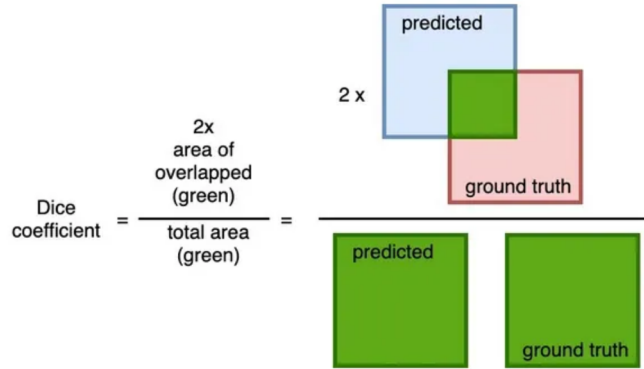


Figure 4.4: Visual explanation of the Dice coefficient calculation. The Dice coefficient is computed as the ratio of twice the area of overlap (green) to the total area of both the predicted segmentation and ground truth. Image adapted from [Nev23].

The Dice loss is derived from the Dice coefficient (also known as F1 score), which measures the overlap between the predicted segmentation and the ground truth. The Dice coefficient is defined as:

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|} \quad (4.1)$$

where X is the predicted set and Y is the ground truth set.

4.6 Hausdorff Distance (95th Percentile)

While the Dice coefficient measures volumetric overlap, the Hausdorff Distance (HD) evaluates the boundary similarity between the predicted segmentation (X) and the ground truth (Y). It is a measure of how far two subsets of a metric space are from each other. Conceptually, it represents the greatest of all the distances from a point in one set to the closest point in the other set.

The standard Hausdorff Distance is defined as:

$$H(X, Y) = \max \left(\sup_{x \in X} \inf_{y \in Y} d(x, y), \sup_{y \in Y} \inf_{x \in X} d(x, y) \right) \quad (4.2)$$

where $d(x, y)$ is the Euclidean distance between points x and y .

A major drawback of this formulation is its sensitivity to outliers. A single misclassified voxel far from the ground truth can dramatically increase the HD value. To address this, the 95th percentile Hausdorff Distance (HD95) is commonly used. Instead of the maximum distance (sup), it computes the 95th percentile of the distances, making the metric more robust to small segmentation errors and outliers. A lower HD95 value indicates a better alignment of the predicted and ground truth boundaries.

4.7 Absolute Lesion Count Difference

The Absolute Lesion Count Difference (ALCD) is an object-level metric that assesses whether the model predicts the correct number of distinct lesions. It is calculated by:

Performing 3D connected component analysis on both the predicted mask and the ground truth mask to identify and count the number of separate objects (lesions).

Calculating the absolute difference between these two counts:

$$ALCD = |\text{Count}_{\text{predicted}} - \text{Count}_{\text{ground_truth}}| \quad (4.3)$$

This metric is clinically relevant as it penalizes models that incorrectly merge distinct lesions into one or split a single large lesion into multiple smaller ones. A lower ALCD score is better, with a score of 0 indicating a perfect match in the number of detected lesions.

4.8 Lesion F1 Score

The Lesion F1 Score is another object-level metric that evaluates the model's ability to detect individual lesions correctly, moving beyond simple voxel-wise comparison. It is

calculated based on the concepts of True Positives (TP), False Positives (FP), and False Negatives (FN) at the lesion level:

True Positive (TP): A predicted lesion that overlaps with a ground truth lesion.

False Positive (FP): A predicted lesion that does not overlap with any ground truth lesion.

False Negative (FN): A ground truth lesion that is not overlapped by any predicted lesion.

From these counts, Precision and Recall are calculated:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (4.4)$$

The Lesion F1 score is the harmonic mean of this lesion-level Precision and Recall:

$$\text{Lesion F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (4.5)$$

This metric provides a balanced assessment of detection accuracy, penalizing both missed lesions and spurious detections. A score of 1 indicates perfect detection, while a score of 0 indicates no correct detections.

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